EXPERIMENT 15

College Security Management System UML Diagrams

Aim

To design the College Security Management System using UML diagrams in Umbrello UML Modeller, including Class, Use Case, and Activity diagrams.

Procedure

Create a new UML project in Umbrello UML Modeller.

Import C++ source files to generate the Admin, CampusLocation, DutySchedule, and SecurityPersonnel classes.

Add the Leave class with its attributes and operations.

Arrange the classes and connect them using association relationships.

Create a Use Case Diagram with Admin, Security Personnel, College Management, and Student actors.

Add the use cases Assign Duty, View Duty Schedule, Apply Leave, Approve Leave, Calculate Salary, and Generate Reports.

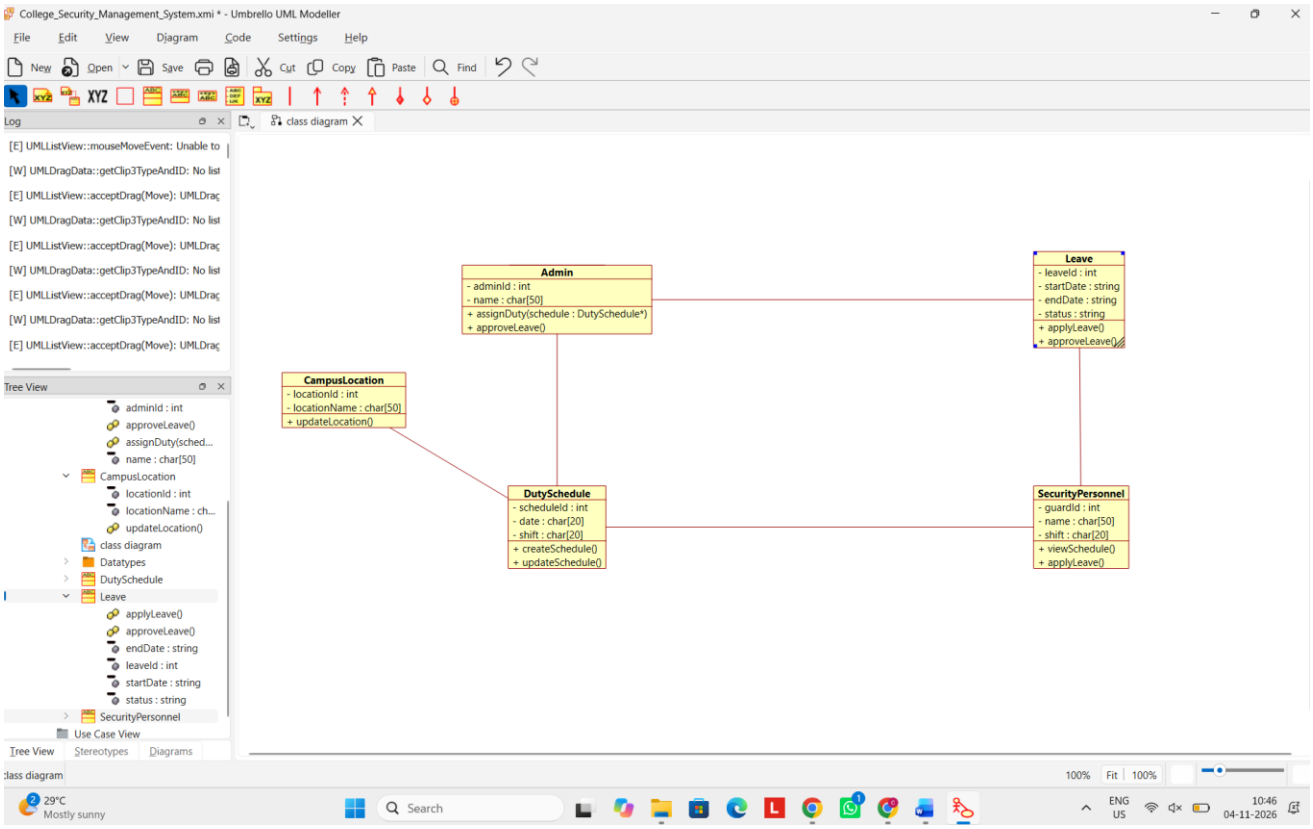
Connect the actors to their corresponding use cases using association relationships.

Create an Activity Diagram with the initial node, activity nodes, control flows, and final node.

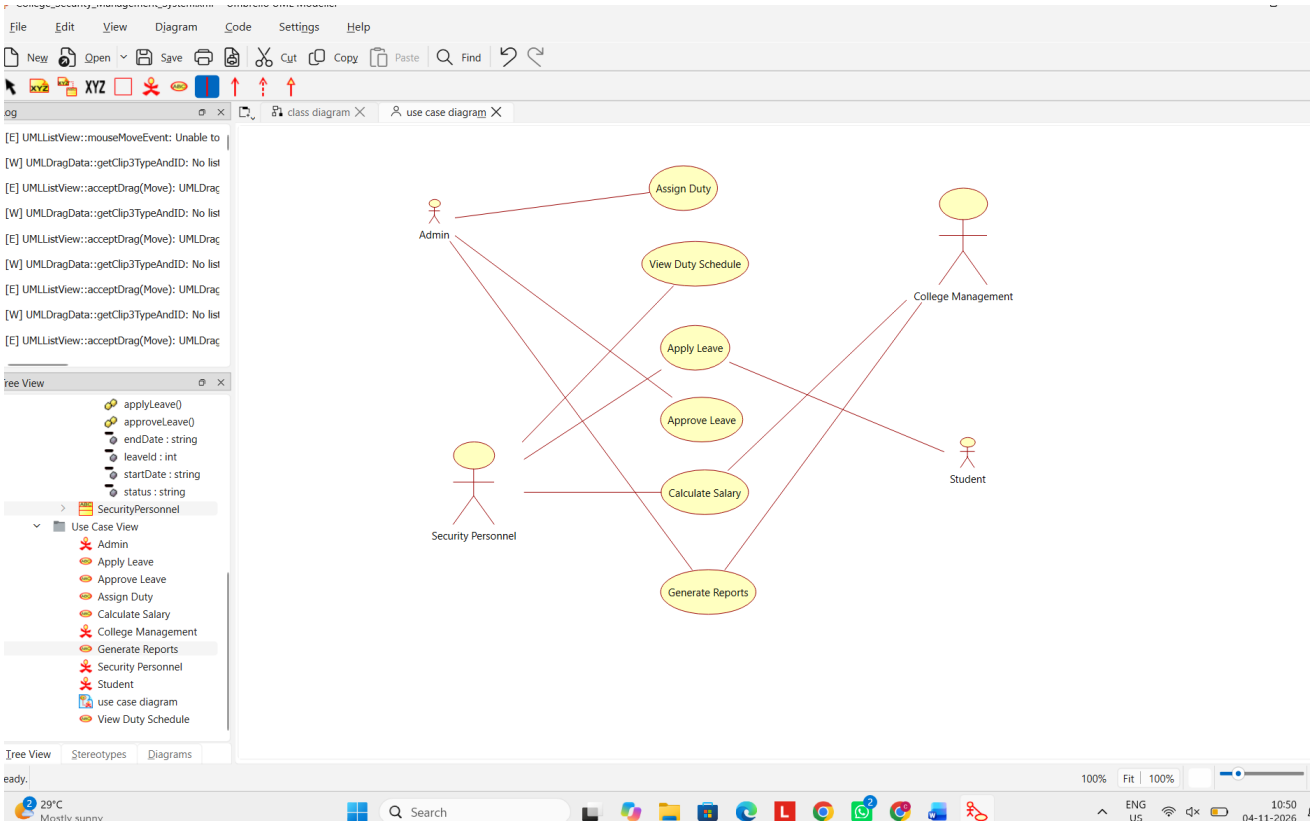
Save the UML project.

Output

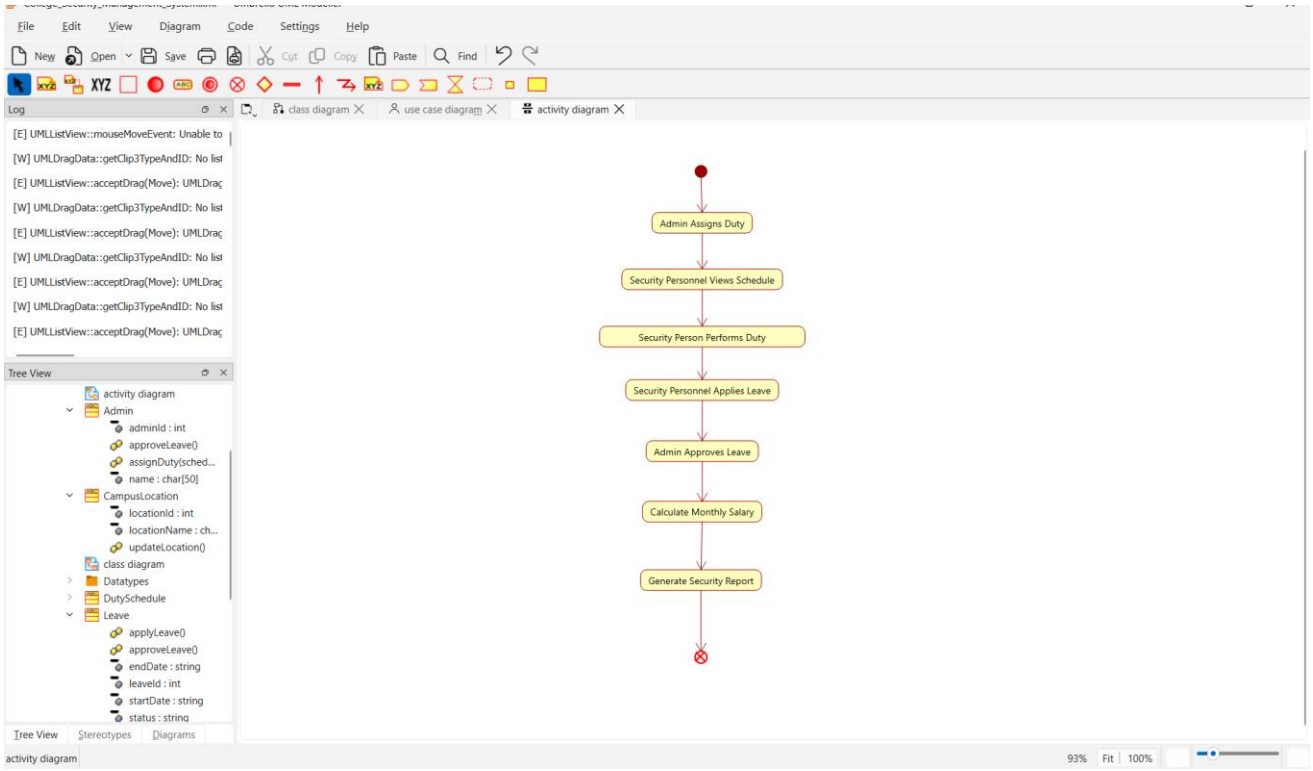
Class Diagram


College_Security_Management_System.xmi * - Umbrello UML Modeller
File Edit View Diagram Code Settings Help
New Open Save Cut Copy Paste Find
Log
class diagram
[E] UMLListView::mouseMoveEvent: Unable to
[W] UMLDragData::getClip3TypeAndID: No list
[E] UMLListView::acceptDrag(Move): UMLDrag
[W] UMLDragData::getClip3TypeAndID: No list
[E] UMLListView::acceptDrag(Move): UMLDrag
[W] UMLDragData::getClip3TypeAndID: No list
[E] UMLListView::acceptDrag(Move): UMLDrag
[W] UMLDragData::getClip3TypeAndID: No list
[E] UMLListView::acceptDrag(Move): UMLDrag
Tree View
adminId : int
approveLeave()
assignDuty(sched...
name : char[50]
CampusLocation
locationId : int
locationName : ch...
updateLocation()
class diagram
Datatypes
DutySchedule
Leave
applyLeave()
approveLeave()
endDate : string
leaveId : int
startDate : string
status : string
SecurityPersonnel
Use Case View
Tree View Stereotypes Diagrams
Admin
- adminId : int
- name : char[50]
+ assignDuty(schedule : DutySchedule*)
+ approveLeave()
Leave
- leaveId : int
- startDate : string
- endDate : string
- status : string
+ applyLeave()
+ approveLeave()
CampusLocation
- locationId : int
- locationName : char[50]
+ updateLocation()
DutySchedule
- scheduleId : int
- date : char[20]
- shift : char[20]
+ createSchedule()
+ updateSchedule()
SecurityPersonnel
- guardId : int
- name : char[50]
- shift : char[20]
+ viewSchedule()
+ applyLeave()
class diagram
100% Fit 100%
29°C
Mostly sunny
Search
ENG
US
10:46
04-11-2026

Use Case Diagram



Activity Diagram



Result

The Class, Use Case, and Activity diagrams for the College Security Management System were successfully created using Umbrello UML Modeller.